

Parv Kapoor

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Research statement: My research focuses on developing new machine learning techniques for safe robot learning combining data-driven approaches (foundation models, imitation learning, reinforcement learning) with classical control techniques (control barrier functions, optimization-based control, reactive synthesis)

Education

Carnegie Mellon University

Ph.D. in Software Engineering | Advisors: Dr. Eunsuk Kang and Dr. Sebastian Scherer

Pittsburgh, PA

May 2026 (expected)

Carnegie Mellon University

Masters in Software Engineering | TA: 10-623 Generative AI, 17-723 Designing large scale systems

Pittsburgh, PA

December 2025 (expected)

Manipal Institute of Technology

B.Tech. in Computer Science Engineering

Manipal, India

August 2020

Skills

Languages Python, C, C++, CUDA, Alloy

Tools and Libraries PyTorch, JAX, ROS 2, Docker, PyTorch Lightning, Isaac Sim

Ph.D. Research

AirLab, Carnegie Mellon University

PI: Sebastian Scherer

Pittsburgh, PA

August 2021 – Ongoing

- [SafeDec](#): Developed constrained decoding frameworks that inject safety constraints into robot foundation models at inference time, validated on object-navigation tasks in AI2-THOR environments in collaboration with Carnegie Bosch Institute
- [STLCG++](#) [RA-L'25]: Built a PyTorch/JAX library integrating differentiable temporal logic into robot learning, delivering 100× speed-ups and outperforming baselines for trajectory optimization and diffusion-policy case studies
- [ViSafe](#) [RSS'25]: Developed vision-based control barrier functions for UAV safety, validated via sim-to-real transfer from Isaac Sim to field, achieving 71% improvement over baselines across 70+ flight hours
- [MCSTL](#) [ICRA'23]: Trained real-world aviation policies using guided imitation learning and online tree search, outperforming baselines by 60%

Software Design and Analysis Lab, Carnegie Mellon University

PI: Eunsuk Kang

Pittsburgh, PA

August 2021 – Ongoing

- [ETL](#): Pretrained latent world models and coupled them with foundation-model vision encoders (DINOv2, CLIP) and MPPI control, enabling robots to plan and act safely through latent-space reasoning in Habitat environments
- [RobRL](#) [FM'24]: Designed stochastic optimization methods to analyze reinforcement learning policy robustness for autonomous driving under actuation and environmental deviations (e.g., steering, friction, and sensor noise) in collaboration with Toyota Motor North America
- [ATLAS](#) [ICSE'25]: Built a framework for learning formal behavior rules from robot demonstrations using MaxSAT for structured policy synthesis
- [STLINC](#) [NFM'24]: Devised requirement decomposition for integrated task and motion planning, reducing solve time by 65%

Experience

Microsoft Research

Machine Learning Research Intern | PI: Dr. John Langford and Dr. Siddhartha Sen

New York, NY

May 2025 – August 2025

- Built [Matryoshka Belief State Transformer \(MatBST\)](#) architecture with hierarchical latent planning, enabling fast long-horizon reasoning in LLMs
- Optimized large-scale pretraining and fine tuning with PyTorch FSDP and custom sharding strategies for large language models
- Evaluated MatBST on infilling tasks that test temporal reasoning and belief-state consistency over long horizons

General Robotics

Student Researcher | PIs: Dr. Jonathan Huang and Dr. Ashish Kapoor

Seattle, WA

September 2024 - May 2025

- Developed pipelines for procedural synthetic data generation for AI2-THOR, Habitat and Unreal Engine environments
- Pretrained multi-modal robot foundation models via large-scale behavior cloning for SOTA mobile manipulation
- Developed aligned robotics foundation models using parameter-efficient and reinforcement fine-tuning for cross-task generalization

General Robotics

Research Intern | PIs: Dr. Sai Vemprala and Dr. Ashish Kapoor

Seattle, WA

May 2024 - August 2024

- [PASTEL](#) (RSS '24): Trained instruction-conditioned robot policies with cross-attention transformers, outperforming baselines by 74.3%
- Contributing to the development of the [AirGen](#) simulation platform for aerial and ground robots
- Built a foundation model field deployment pipeline for diverse robots (Unitree Go2, DJI Robomaster) using domain adaptation

Cyber Physical Systems Lab, University of Southern California

Research Intern | PI: Dr. Jyotirmoy Vinay Deshmukh

Los Angeles, CA

January 2020 - January 2021

- Developed novel model-based reinforcement learning algorithms for safe policy training from signal temporal logic specifications
- Implemented efficient model-free algorithms (TRPO, A3C, PPO) in PyTorch with unique STL-based reward design, achieving 82% higher specification satisfaction compared to baseline RL policies
- Engineered in-house simulation environments for algorithm benchmarking employing CARLA, AirSim, and Gazebo

Visual Computing Group, Cardiff University

Research Intern | PI: Dr. David Marshall

Cardiff, U.K.

May 2019 - July 2019

- Constructed a safe trajectory prediction system for visually impaired individuals using ZED stereo camera
- Implemented and trained a 2-stream CNN in TensorFlow on human walking data for forecasting ego agent camera movement
- Improved CNN accuracy in low-data regimes through neuro-inspired data augmentation

Preprints and Publications

Scaling Long-Horizon Reasoning via Hierarchical Matryoshka Belief State Transformers

P. Kapoor, S. Sen, J. Langford

- Under Submission

Constrained Decoding for Robotics Foundation Models

P. Kapoor, A. Ganlath, C. Liu, S. Scherer, E. Kang

- Under Submission [\[arxiv\]](#)

Pretrained Embeddings as a Behavior Specification Mechanism

P. Kapoor, A. Hammer, A. Kapoor, K. Leung, E. Kang

- Under Submission [\[arxiv\]](#)

ViSafe: Vision-enabled Safety for High-speed Detect and Avoid

P. Kapoor, I. Higgins, N. V. Keetha, J. Patrikar I. Cisneros, Z. Ye, Y. He, Y. Hu, C. Liu, E. Kang, S. Scherer

- Robotics Science and Systems (RSS) 2025 [\[arxiv\]](#)

STLCG++: A Masking Approach for Differentiable Signal Temporal Logic Specification

P. Kapoor, K. Mizuta, E. Kang, K. Leung

- IEEE Robotics and Automation Letters (RA-L), presented at IEEE International Conference on Robotics and Automation (ICRA) 2026 [\[arxiv\]](#)

Logically Constrained Robotics Transformers for Enhanced Perception Action Planning

P. Kapoor, S. Vemprala, A. Kapoor

- Robotics Science and Systems (RSS): Towards Safe Autonomy 2024 [\[arxiv\]](#)

Constrained LTL Specification Learning from Examples

C. Zhang*, P. Kapoor*, I. Dardik, A. Cui, R.M. Goes, D. Garlan, E. Kang

- IEEE International Conference on Software Engineering (ICSE) 2025 [\[arxiv\]](#)

Analyzing Tolerance of Reinforcement Learning Controllers against Deviations in Cyber Physical Systems

C. Zhang*, P. Kapoor*, R.M. Goes, D. Garlan, E. Kang, A. Ganlath, S. Mishra, N. Ammar

- Formal Methods (FM) 2024 [\[arxiv\]](#)

Safe Planning through Incremental Decomposition of Signal Temporal Logic

P. Kapoor, R.M. Goes and E. Kang

- Nasa Formal Methods (NFM) 2024 [\[arxiv\]](#)

Follow The Rules: Online Signal Temporal Logic Tree Search for Guided Imitation Learning in Stochastic Domains

J. Patrikar, J. Aloor, P. Kapoor, S. Scherer and J. Oh

- IEEE International Conference on Robotics and Automation (ICRA) 2023 [\[arxiv\]](#)

Challenges in Close-Proximity Safe and Seamless Operation of Manned and Unmanned Aircraft in Shared Airspace

J. Patrikar, J. Dantas, S. Ghosh, P. Kapoor et al

- IEEE International Conference on Robotics and Automation (ICRA) 2022 [\[arxiv\]](#)

Model-based Reinforcement Learning from Signal Temporal Logic Specifications

P. Kapoor, A. Balakrishnan, J. V. Deshmukh

- [\[arxiv\]](#)